

1. Let be the subsets of $\mathbb{R}$ defined by

$$A = \left\{ \frac{1}{2} - \frac{n}{2n+1}, n \in \mathbb{N} \cup \{0\} \right\}, B = \mathbb{N}, C =]0, 11] \cup \{1, 2\}.$$

- Determine the accumulation, isolated points of each.
 - Deduce the closure of each.
2. Let A be a subset of $\mathbb{R}$. Show that:
- A is open iff it's neighborhood of all its points.
 - A is closed iff it contains all its accumulation points.
3. • Write the following complex numbers using the trigonometric form:
 $z_1 = 3 + 3i, Z_2 = -1 - i\sqrt{3}, z_3 = \exp i\theta + \exp 2i\theta$.
- Write the following complex numbers using the algebraic form:
 $z_1 = \frac{3+6i}{3-4i}, Z_2 = \left(\frac{1+i}{2-i}\right)^2, z_3 = \frac{(1+i)^9}{(1-i)^7}$.
4. Let $x \in \mathbb{R}$ and $u \in \mathbb{C}$ such that $|u| = 1$. Show that $|x - u| = |1 - xu|$ without using the algebraic form of u .
5. Show, without using the algebraic form, if u and v are two complex numbers of moduli equal to 1, then $\frac{u+v}{1+uv}$ is real, and $\frac{u-v}{u-v}$ is imaginary. – What about $\frac{u+v}{1-uv}$?
6. Let z_1, z_2 and z_3 be three complex numbers with moduli equal to 1. Compare the moduli of
- $$z_1 + z_2 + z_3 \quad \text{and} \quad z_1z_2 + z_2z_3 + z_1z_3.$$
7. Show that every complex number of modulus equal to 1 can be written in the form $\frac{1+ix}{1-ix}$, where x is real.
8. Evaluate the modulus and the argument of $z = \frac{1}{1+i \tan \alpha}$.
9. Let z_1 and z_2 be any given complex numbers.
- (a) Verify the following parallelogram law (also called the parallelogram identity):

$$|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2(|z_1|^2 + |z_2|^2)$$

- (b) Let u be a complex number such that $u^2 = z_1.z_2$. Prove the following identity:

$$|z_1| + |z_2| = \left| \frac{z_1 + z_2}{2} + u \right| + \left| \frac{z_1 + z_2}{2} - u \right|$$

10. Prove by using complex numbers, that for all real numbers a, b, c , we have

$$\sqrt{2}|a + b + c| \leq \sqrt{a^2 + b^2} + \sqrt{b^2 + c^2} + \sqrt{c^2 + a^2}.$$

11. find the exponential form of the following complex numbers:

$$z_1 = 1 + \cos \varphi - i \sin \varphi \text{ and } z_2 = \frac{\cos x - i \sin x}{\sin x - i \cos x}.$$

12. Ptolemy's inequality:

(a) Let u and v be nonzero complex numbers. Show that :

$$\left| \frac{u}{|u|^2} - \frac{v}{|v|^2} \right| = \frac{|u - v|}{|u| |v|}.$$

(b) Let u, v and w be complex numbers. Prove the following Ptolemy's inequality:

$$|u - v| |w| \leq |u - w| |v| + |v - w| |u|$$

13. Let $w = \exp\{\frac{2\pi i}{n}\}$. Prove that :

$$1) \ 1 + 2w + 3w^2 + \dots + nw^{n-1} = \frac{n}{w - 1}.$$

$$2) \ (z - w)(z - w^2) \dots (z - w^{n-1}) = 1 + z + \dots + z^{n-1}.$$

14. Calculate $\left(\frac{1 + i\sqrt{3}}{1 - i} \right)^{60}$.

15. Solve in $\mathbb{C}$, the following equations

$$z^n = \bar{z}, \quad n \in \mathbb{N}.$$

$$(z + i)^n = (z - i)^n, \quad n \in \mathbb{N}, \ n \geq 2.$$

$$1 + 2z + 2z^2 + \dots + 2z^{n-1} + z^n = 0, \quad n \in \mathbb{N}^*$$

$$(z^2 + z + 1)^2 + 1 = 0.$$